

# Donghong Cai

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🌐 <https://ilikevegetable.github.io>

## EDUCATION

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### Washington University in St. Louis

*Ph.D. in Computer Science*

- **Advisor:** Dr. Yixin Chen

St. Louis, USA

*Aug 2024 – present*

### Zhejiang University

*M.S. in Software Engineering; GPA: 3.84/4.00*

- **Research Directions:** Time Series and Graph Data Mining
- **Advisor:** Dr. Yang Yang

Hangzhou, China

*Sep 2020 – Mar 2023*

### Huazhong University of Science and Technology

*B.S. in Software Engineering; GPA: 3.67/4.00*

Wuhan, China

*Sep 2016 – Jun 2020*

## INDUSTRY EXPERIENCE

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### Alibaba Group

*Machine Learning Engineer, International E-commerce Seller Team*

Shenzhen, China

*Apr 2023 – Jul 2024*

- Developed recommendation algorithms for seller promotional tools (e.g., Flexi Combo, New Buyer Voucher), improving merchants' GMV by 8.28% and Units Ordered by 6.55% on average.
- Designed a novel cross-platform seller matching algorithm to identify potentially identical stores on competing e-commerce platforms.
- Built a transformer-based time series forecasting model to predict monthly GMV and order volume, improving prediction accuracy by 21.7% and 36.8% over manual heuristics.
- Developed an LLM-based seller assistant using a hybrid two-stage retrieval framework, combining QA-pair semantic matching for frequent queries with Retrieval-Augmented Generation (RAG) over large-scale multimodal data sources (databases, documents, etc.).

## RESEARCH EXPERIENCE

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### Free-form Tabular Data Generation

*Graduate Research Assistant*

Washington University in St. Louis

*Aug 2025 – Jan 2026*

- Proposed **TabDLM**, a unified diffusion framework for synthetic tabular data generation that jointly models numerical, categorical, and free-form textual features within a masked diffusion language model (MDLM).
- Designed a hybrid numerical-language diffusion architecture with specialized numerical tokenization, enabling continuous diffusion for numerical attributes and masked language diffusion for textual fields to capture cross-modality dependencies.
- Conducted extensive experiments on synthetic and real-world tabular benchmarks, demonstrating superior distribution fidelity, cross-field consistency, and downstream predictive utility compared with state-of-the-art diffusion- and LLM-based baselines.

### Pre-training Framework for Brain Signals

*Graduate Research Assistant*

Zhejiang University

*Jun 2021 – Feb 2022*

- Designed a generalized self-supervised learning framework **MBrain** consisting of three well-designed tasks, which can be applied to pre-train both EEG and SEEG brain signals.
- Proposed multi-channel CPC (Contrastive Predictive Coding) and theoretically proved that optimizing the goal of multi-channel CPC can lead to a better predictive representation. Based on the multi-channel CPC, three self-supervised learning tasks were designed to explicitly capture the spatial and temporal correlations of brain signals to learn informative representations for downstream tasks.
- Validated the superior effectiveness and clinical value of the proposed framework through extensive experiments of seizure detection on large-scale real-world EEG and SEEG datasets. In subject independent seizure detection experiment which meet practical clinical needs, **MBrain** outperforms all baselines on  $F_2$ -score with an increase of **9.23%** and **27.83%** in EEG and SEEG datasets.

# Epileptic Wave Detection Using Hierarchical Graph Diffusion Learning

Graduate Research Assistant

Zhejiang University

Mar 2021 – May 2021

- Formulated and studied the epileptic wave detection problem for SEEG data using an automatic end-to-end data-driven method.
- Proposed **BrainNet** to simultaneously learn the dynamic diffusion graphs and model the brain wave diffusion patterns thereon in a hierarchical fashion to achieve accurate epileptic wave detection under conditions of imbalanced labels and severe noise.
- Conducted comprehensive experiments on a large-scale real-world SEEG dataset across multiple patients. The experimental results validated the effectiveness of *BrainNet* on epileptic wave detection and its superiority in capturing the diffusion process. In the channel-level epileptic wave detection task, *BrainNet* outperforms all baselines on  $F_2$ -score with an increase of **36.66%**.

## PUBLICATIONS

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1. **Donghong Cai**, Jiarui Feng, Yanbo Wang, Da Zheng, Yixin Chen, Muhan Zhang. TabDLM: Free-Form Tabular Data Generation via Joint Numerical–Language Diffusion. Preprint.
2. Jiarui Feng, **Donghong Cai**, Yixin Chen, Muhan Zhang. GRIP: In-Parameter Graph Reasoning through Fine-Tuning Large Language Models. Preprint.
3. **Donghong Cai**, Junru Chen, Yang Yang, Teng Liu, and Yafeng Li. 2023. MBrain: A Multi-channel Self-Supervised Learning Framework for Brain Signals. *In Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '23)*.

## HONORS & AWARDS

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|-----------------------------------------------------------------------|--------------------|
| Award of Honor for Graduate                                           | Dec 2021           |
| Outstanding Graduate of Huazhong University of Science and Technology | Jun 2020           |
| Academic Excellence Scholarship                                       | Sep 2018           |
| Individual Scholarship                                                | Sep 2018, Sep 2019 |

## LANGUAGES & SKILLS

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**Languages:** Chinese (Native), English (Fluent)

**Programming:** C, C++, Java, JavaScript, Python, MATLAB, MySQL

**Professional Skills:** SKlearn, PyTorch, Markdown, LaTeX, Shell Script

**Interests:** Basketball, Football, Hiking, Photography